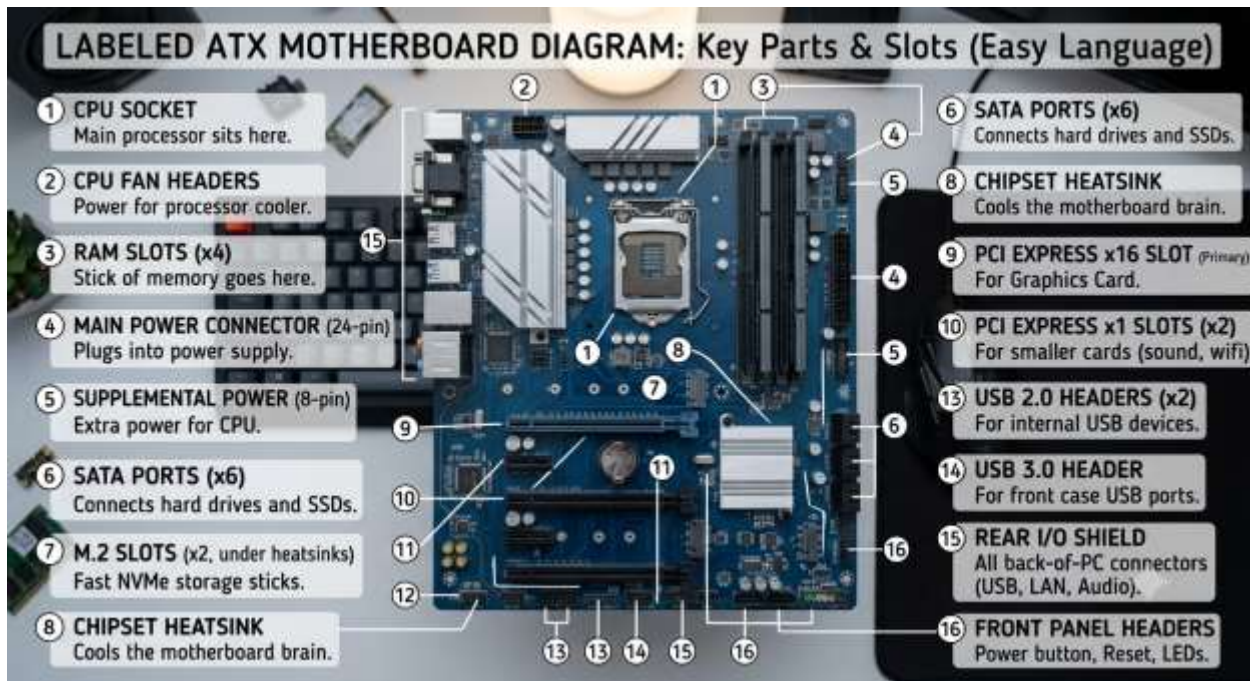


HARDWARE

1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).

Ans. The labeled diagram of ATX motherboard and there slots.



RAM slots = this is physical slot in mother board where we plug the RAM in the slot.

1. **PCIe slots** = this is the slot in mother board where we plug the Graphic card for better performance.
2. **SATA ports** = this port is for connecting the storage drive.
3. **USB headers** = this ports is in the mother board to connect the device.
4. **CPU socket** = this is slot in our mother board where we put the processor.
5. **Power connectors** = this give the electricity to start the pc
6. **Front panel** = in this where the button, ports, and lights in the front of cabinet.
7. **Chipset** = this is the traffic controller of mother board

2.. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

Ans. **1. AMD AM5 (LGA 1718)**

*This is make in 2022 this is mainly for Ryzen consumer for the desktop processor

Compatible CPU Model: AMD Ryzen 7 9800X3D

2. Intel LGA 1700

*is simply the name for that specific socket where a certain type of Intel computer brain can fit.

Compatible CPU Model: Intel Core i9-14900K

3. Intel LGA 1851

*It's a specific socket on the motherboard designed to hold and power only the newest Intel CPUs.

Compatible CPU Model: Intel Core Ultra 9 285K

3.Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

Ans. The CPU = this is the most power full chip in our computer and the primary function is **Thinking** and **Math**.

* it was very smart and faster brain that make the decisions.

The Chipset = in the chipset there is the set of smaller chips that fixed in mother board.

*the chipset primary work is **Communication** and **Logistics**.

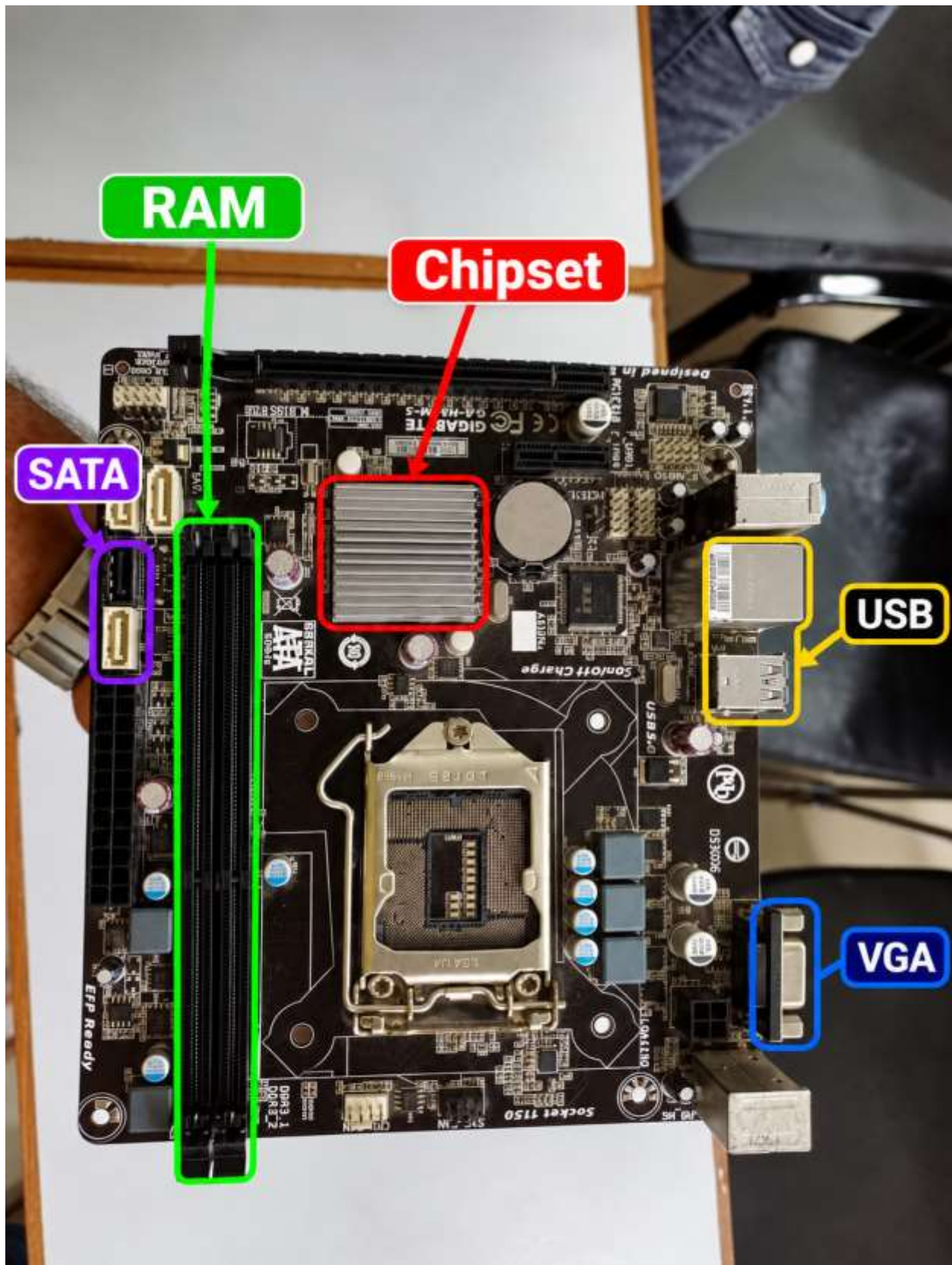
How they work together – to open Spotify

*When we click on the icon app Spotify the request can pass to chipset and then the chipset give the alert to CPU that you want to run the app.

*then the CPU tells the chipset to take the file from storage device and than the chipset take the data and the CPU processer can play the and play the music.

4.Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

Ans.



1. **SATA ports** = this is the ports in our motherboard use for storage device
2. **VGA ports** = the connector for monitors and projectors
3. **USB ports** = a port that connect device to the computer
4. **RAM** = this slot is in our motherboard where we put the RAM stick for use the short-term memory
5. **Chipset** = this is the set of chips on the motherboard to manage the communication between the CPU, RAM, Storage and other.